

1. In an alloy, zinc and copper are in the ratio of 1 : 2. In the second alloy, the same elements are in the ratio 2 : 3. If these two alloys be mixed to form a new alloy in which two elements are in the ratio 5 : 8, the ratio of these two alloy in the new alloy is:
(1) 3 : 10 (2) 3 : 7 (3) 10 : 3 (4) 7 : 3

2. A Jar contained a mixture of two liquids *A* and *B* in the ratio 4 : 1. When 10 litres of the mixture was taken out and 10 litres of liquid *B* was poured in the Jar, this ratio became 2 : 3. The quantity of liquid *A* contained in the Jar initially was:
(1) 4 litres (2) 8 litres (3) 16 litres (4) 32 litres

3. In a mixture of 75 litres, the ratio of milk to water is 2 : 1. The amount of water to be further added to the mixture so as to make the ratio of the milk to water 1 : 2 will be:
(1) 45 litres (2) 60 litres (3) 75 litres (4) 80 litres

4. The ratio in which two sugar solutions of the concentrations 15% and 40% are to be mixed to get a solution of concentration 30% is _____
(1) 2 : 3 (2) 3 : 2 (3) 8 : 9 (4) 9 : 8

5. 300 gm of sugar solution has 40% sugar in it. How much sugar should be added to make it 50% in the solution?
(1) 40 gm (2) 50 gm (3) 60 gm (4) 80 gm

6. In a zoo, there are rabbits and pigeons. If the heads are counted, there are 200 and if legs are counted, there are 580. How many pigeons are there?
(1) 105 (2) 110 (3) 115 (4) 120
7. A man has 60 pens. He sells some of these at a profit of 12% and the rest at 8% loss. On the whole, he gets a profit of 11%. How many pens were sold at 12% profit?
(1) 47 (2) 52 (3) 55 (4) 57
8. The ratio of milk and water in 66 liters of adulterated milk is 5 : 1. Water is added to it to make the ratio 5 : 3. The quantity of water added is:
(1) 20 lt. (2) 22 lt. (3) 24 lt. (4) 28 lt.
9. A sum of Rs. 41 was divided among 50 children. Each boy gets 90 paise and each girl 61 paise. The number of boy is:
(1) 32 (2) 34 (3) 36 (4) 38
10. How much tea at Rs. 4 a kg should be added to 15 kg tea at Rs. 10 a kg, so that the mixture be worth Rs. 6.50 a kg?
(1) 15 kg (2) 35 kg (3) 25 kg (4) 21 kg

11. A vessel of 80 litre is filled with milk and water. 70% of milk and 30% of water are taken out of the vessel. It is found that the vessel is empty by 55%. Find the initial quantity of milk and water.
(1) 20 lt, 30 lt (2) 30 lt, 50 lt (3) 40 lt, 40 lt (4) 50 lt, 30 lt
12. In what proportion water be mixed with milk to gain 12.5% by selling it at last price?
(1) 3 : 5 (2) 1 : 8 (3) 2 : 7 (4) 1 : 9
13. In what ratio should freely available water to be mixed with soda worth Rs. 60 per litre, so that after selling the mixture at Rs. 50 per litre the profit will be 25%?
(1) 1 : 2 (2) 2 : 3 (3) 3 : 4 (4) 4 : 5
14. A mixture of water and milk contains 80% milk. In 50 litres of such a mixture, how many litres of water is required to increase the percentage of water of 50%?
(1) 20 (2) 15 (3) 30 (4) None of these
15. The average weight of a class of 40 students is 30 kg and the average weight of a class of 20 students is 15 kg. Find the average weight of both the classes combined.
(1) 20 (2) 25 (3) 17.5 (4) 15

16. In 80 litres mixture of milk and water the ratio of milk and water 3 : 2. How many litres milk should be added to the mixture so that milk may become double of the water.
(1) 16 litre (2) 12 litre (3) 20 litre (4) 14 litre
17. A bus agency has 108 buses. He sold some bus at 9% profit and rest at 36% profit. Thus he gains 17% on the sale of all his buses. The no. of buses sold at 36% profit is:
(1) 25 (2) 32 (3) 35 (4) 75
18. Two vessels A and B contain milk and water in the ratio 8 : 5 and 5 : 2 respectively. The ratio in which these two mixtures be mixed to get a new mixture containing $69\frac{3}{13}\%$ milk _____
(1) 3 : 5 (2) 5 : 2 (3) 5 : 7 (4) 2 : 7
19. A container contains 60 kg of milk. From this container 6 kg of milk was taken out and replaced by water. This process was repeated further two times. The amount of milk left in the container is _____
(1) 34.24 kg (2) 39.64 kg (3) 43.74 kg (4) 4.76 kg
20. 7 kg of tea costing Rs.280 per kg is mixed with 9 kg of tea costing Rs. 240 per kg. The average price per kg of the mixed tea is _____
(1) Rs. 255.80 (2) Rs.257.50 (3) Rs. 267.20 (4) Rs. 267.50

21. Two types of tea costing Rs.180 / kg and Rs. 280 / kg. In what ratio should these be mixed so that the obtained mixture sold at Rs.320 / kg to earn a profit of 20% is _____
(1) 3 : 13 (2) 4 : 13 (3) 1 : 13 (4) 2 : 13
22. Three glasses of equal volume contain acid mixed with water. The ratio of acid and water are 2 : 3, 3 : 4 and 4 : 5 respectively. Contents of these glasses are poured into a large vessel. The ratio of acid and water in the large vessel is _____
(1) 417 : 564 (2) 401 : 544 (3) 407 : 560 (4) 411 : 540
23. The proportion of acid and water in three samples is 2 : 1, 3 : 2 and 5 : 3. A mixture containing equal quantities of all three samples is made. The ratio of water and acid in the mixture is _____
(1) 120 : 133 (2) 227 : 133 (3) 227 : 120 (4) 133 : 227
24. A gold smith has two qualities of gold, one of 12 carats and another of 16 carats purity. In what proportion should he mix both to make an ornament of 15 carats purity?
(1) 1 : 3 (2) 2 : 3 (3) 3 : 2 (4) 3 : 1
25. Lala has lent some money to Arun at 5% per annum and Bhatia at 8% per annum. At the end of the year, he has gained an overall interest at the rate 6%. On what ratio has he lent the money to Arun and Bhatia?
(1) 2 : 1 (2) 1 : 2 (3) 3 : 2 (4) 3 : 1

26. The acid and water in two vessels A and B are in the ratio 4 : 3 and 2 : 3. In what ratio should the liquid in both the vessel be mixed to obtain a new mixture in vessel C containing half acid and half water?
(1) 7 : 5 (2) 5 : 7 (3) 7 : 3 (4) 5 : 3
27. Two vessel A and B contain milk and water mixed in the ratio 8 : 5 and 5 : 2 respectively. The ratio in which these two mixture be mixed to get a new mixture containing $69\frac{3}{13}\%$ milk is _____
(1) 3 : 5 (2) 5 : 2 (3) 5 : 7 (4) 2 : 7
28. A can is full of a mixture of two liquids A and B in the ratio of 7 : 5. When 9 litres of mixture are drawn off from the can and replaced by the same quantity of liquid B, the ratio of A and B in the can becomes 7 : 9. The capacity of the can is _____
(1) 21 litres (2) 20 litres (3) 10litres (4) 36 litres
29. A bartender stole champagne from a bottle that contained 50% of spirit and he replaced what he had stolen with champagne having 20% spirit. The bottle then contained only 25% spirit. How much of the bottle did he steal?
(1) 80% (2) 83.33% (3) 85.71% (4) 88.88%
30. In what proportion must a grocer mix sugar at Rs. 12 a kg and Rs.7 a kg sugar as to make a mixture worth Rs. 8 a kg?
(1) 7 : 12 (2) 1 : 4 (3) 2 : 3 (4) 12 : 7